Binhua Huang

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PROJECTS

2022.3 - An Optical Wearable Device for ASL Recognition (First author)

- The experiment was completed. The classification accuracy for American Sign Recognition reached 99.8% with KNN.

2021.11 - Liquid Volume Estimation by Tactile Sensor Using Machine Learning (First author)

- The paper was finished. Now in the revision process. The classification accuracy for liquid volume estimation reached 99.78% with CNN.

2021.10- An IoT Based Emergency Cardiac Death Risk Rescue Alert System (First author)

- The paper was finished. The other author (Dr. Shafiq Ul Rehman) and the corresponding author (Dr. Selvakumar Manickam) suggest some revisions.

2021.8- Closed-loop Control of a Soft Hand by Tactile Perception (Member)

- My role: The closed-loop control strategy development and coding(python), the circuit design, communication protocol implementation， experiment implement, and data collection.

SKILLS

8Bit,32Bit MCU / FPGA | C/C++/C#/Python/VB | Digital/Analog Circuit Design and PCB Layout | Proteus\Multisim\Matlab\ Quartus | Machine Learning | SAS Testing

EXPERIENCE

VISITING STUDENT (RESEARCH ASSISTANT)

Intelligent Systems and Robot Learning Lab, Shenzhen Institute of Advanced Technology (SIAT), Chinese Academic of Science (CAS), China, August 2021 - Present

- Hardware R&D (Circuit design, Signal system analysis, Sensor development)
- Low-level coding for control, communication, and sensing systems
- Machine learning algorithms optimization and implementation

TEST AUTOMATION DEVELOPER

HQ of Rural Credit Bank, China, February 2019 – February 2020

- Improve efficiency for computing huge data (60TB)
- Automatic data comparison strategies and implement

TECHNICAL SUPPORT ENGINEER

Branch of Rural Credit Bank, China, May 2016 – February 2019

- User data analysis & computer technical support

EDUCATION

M.S. in Computer Science
USM, University of Science, Malaysia
April,2022 (GPA:3.5)

B.E. in Electronic Information Engineering
Longyan University, China
July,2015 (GPA: 79/100, Rank:15%)